Module-Enhanced Slope-Adaptive and Hazard-Aware Hexapod Robotic System for Safe Roof Inspection

Xiayu Zhao, S.M.ASCE[1]; Yizhi Liu, Ph.D., M.ASCE[2]; and Houtan Jebelli, Ph.D., M.ASCE[3]

Abstract: Roof inspection is essential yet perilous in construction, with falls causing 24% of fatal work injuries. While drones allow rapid surveying, they often lack close-range precision, and ground-based robots struggle with pitched, irregular rooftops. To address these challenges, this paper presents a module-enhanced hexapod robot designed for autonomous roof inspection. Our system incorporates three key innovations: (1) a Bézier-curve-based gait algorithm for enhanced stability, (2) a fuzzy-logic slope adaptation controller for effective navigation on inclined surfaces, and (3) an infrared sensor–based edge detection mechanism to prevent hazardous falls. The robotic system employs an 18 degrees-of-freedom six-legged walking platform equipped with an inertial measurement unit for real-time posture adjustment and eight strategically positioned infrared sensors for comprehensive hazard detection. The stability-enhanced gait system utilizes third-degree Bézier curve trajectories to optimize foot placement and minimize body oscillations during locomotion. The fuzzy logic controller dynamically adjusts the robot's posture based on real-time slope measurements, maintaining the center of gravity within the conservative support polygon. The edge detection system combines threshold-based filtering with hierarchical path planning to enable proactive hazard avoidance. Extensive tests—combining virtual simulations with real-world trials on roofs up to a 6/12 pitch (26.6°)—demonstrate 33%–39% improvements in stability metrics compared with conventional hexapod control methods. The system achieved over 90% hazard detection accuracy, while maintaining safe edge clearances of 0.28–0.32 m. Real-world validation confirmed the system's effectiveness across various roof configurations, including challenging structural transitions such as ridge lines and valleys. These results confirm the robot's effectiveness in mitigating fall risks and underscore its potential to advance autonomous operations in challenging, sloped environments, contributing significantly to construction safety and inspection efficiency. **DOI: 10.1061/AOMJAH.AOENG-0088**. *This work is made available under the terms of the Creative Commons Attribution 4.0 International license, https://creativecommons.org/licenses/by/4.0/.*

Practical Applications: This research addresses a critical safety challenge in the construction industry, in which roof inspection work results in hundreds of fatalities annually due to falls from heights. The developed robotic system offers construction companies, insurance agencies, and building maintenance firms a safer alternative to traditional manual roof inspections. Instead of sending workers onto potentially dangerous sloped surfaces, operators can deploy this six-legged robot to conduct detailed inspections, while remaining safely on the ground. The robot can navigate common residential roof slopes, detect damage such as missing shingles or structural defects, and avoid dangerous edges that could cause falls. The system's ability to maintain stability on inclined surfaces up to steep residential roof pitches makes it practical for inspecting the majority of existing buildings. The robot's edge detection capabilities prevent costly accidents and equipment loss, making it economically viable for widespread adoption. As the construction industry faces ongoing labor shortages and increasing safety regulations, this autonomous inspection technology represents a practical solution that can reduce workplace injuries, lower insurance costs, and improve the efficiency of building maintenance operations across residential and commercial properties.

Author keywords: Construction robotics; Autonomous roof inspection; Adaptive control; Module-enhanced robotic system; Hexapod robot.

Introduction

Roof maintenance and inspection are vital for building upkeep, preserving structural integrity, and preventing costly water damage. The roofing industry contributes 56 billion dollars annually to the US economy, accounting for about 10% of the construction budget, with roof replacements dominating over 90% of volume and value (Ridgeline 2023). The economic impact of roof damage is substantial—a survey of 279 manufactured homes affected by hurricanes Irma and Michael revealed that 59% experienced nonstructural damage, while 30% suffered structural damage (Sutley et al. 2020). However, traditional manual inspections expose workers to significant safety risks, with falls from roofs accounting for 24% of all fatal work injuries in the construction industry in 2020 (BLS 2021). In recent years, the construction industry has seen growing interest in automated inspection systems. The advent of construction robots is transforming workflows—robotic arms enhance precision in material handling and assembly (Liu et al. 2025; Ojha et al. 2025), drones conduct aerial surveys providing detailed site data (Panigati et al. 2025; Ren and Jebelli 2025), and autonomous vehicles increase safety and efficiency in ground-based tasks (Ge and Sadhu 2025; Zhao et al. 2024). Robotic systems for building inspection have made significant strides, with

[1]Ph.D. Candidate, Dept. of Civil and Environmental Engineering, Univ. of Illinois Urbana-Champaign, Urbana, IL 61801. Email: xiayuz3@illinois.edu

[2]Assistant Professor, Dept. of Civil and Environmental Engineering, Syracuse Univ., Syracuse, NY 13244. Email: yliu580@syr.edu

[3]Assistant Professor, Dept. of Civil and Environmental Engineering, Univ. of Illinois Urbana-Champaign, Urbana, IL 61801 (corresponding author). Email: hjebelli@illinois.edu

Note. This manuscript was submitted on May 29, 2025; approved on August 23, 2025; published online on October 7, 2025. Discussion period open until March 7, 2026; separate discussions must be submitted for individual papers. This paper is part of the ***ASCE OPEN: Multidisciplinary Journal of Civil Engineering***, ASCE, ISSN 2995-4266.

researchers exploring various locomotion methods suitable for navigating complex architectural structures, while AI-equipped inspection robots can detect maintenance needs more accurately than humans (Bock 2015).

However, many robots face challenges in on-site refined inspection work, limiting practical applicability (Mirjan et al. 2019). Roof complexity poses challenges; visible damage like curling, cracks, and blown-off shingles is just part of the problem, while hidden issues such as loosening, water damage, and blistering complicate inspections (Miura et al. 2020). While unmanned aerial vehicles (UAVs) show promise, they face limitations in close-up inspections, payload capacity, battery life, and regulatory constraints (Munawar et al. 2021). Unmanned ground vehicles (UGVs) offer advantages for close-up inspection with higher payload and longer battery life, making them more suitable for detecting less apparent defects (Zhao and Jebelli 2025a). Among ground robot configurations, hexapod robots offer compelling advantages on inclined surfaces. Their six-legged design provides superior stability through wide support polygons and can precisely place feet to minimize roofing material damage, while maintaining stability through independent leg control (He and Gao 2020). However, considering roof structure complexity with slopes and discontinuous surfaces, there is a lack of UGV robot control systems tailored for safe navigation with slope adaptability and hazard awareness (Wang et al. 2024).

To address this knowledge gap, this study aims to develop a multimodule integrated hexapod robotic system with slope adaptivity and hazard awareness for automated roof inspection. The primary objectives of this research are threefold: (1) designing a mechanism to enhance the hexapod robot's gait stability, while adjusting the posture to adapt the specific slope when walking on inclined roof surfaces, (2) designing a sensing module that allows edge detection and avoidance while walking on the roof, and (3) integrating and comprehensively testing the module-combined system in both simulated and real-world scenarios. The proposed hexapod robot platform incorporates several key enhancements, including a Bézier curve trajectory generator for improved gait stability, a fuzzy logic controller for adaptive posture adjustment on slopes, and an infrared sensor–based edge detection and avoidance system.

This research contributes to the field in several significant ways. First, it presents a novel integration of advanced gait algorithms with adaptive control systems specifically optimized for roof inspection tasks. Second, it demonstrates a comprehensive safety framework that enables autonomous navigation of complex roof structures. Third, it provides detailed validation through both virtual and physical testing, establishing practical guidelines for deploying robotic systems in construction applications.

Related Work

Robotic Systems for Construction and Inspection

The global robotics market is expected to grow from USD 37.81 billion in 2017 to USD 158.21 billion by 2025 at a compound annual growth rate of 19.11% (Fior Markets 2019). The integration of robotic technologies has led to a paradigm shift in building inspection tasks (Liang et al. 2024; Zhao et al. 2025), transitioning from traditionally human-executed tasks to automated platforms (Halder and Afsari 2023; Liu et al. 2021). This transition is driven by the need for enhanced safety, efficiency, and accuracy in construction and inspection processes. Robots are increasingly deployed for structural health monitoring, quality control, and damage assessment, incorporating advanced sensing technologies such as LiDAR, infrared cameras, and high-resolution optical sensors (Lattanzi and Miller 2017; Zhao and Jebelli 2025a).

However, existing robotic inspection systems exhibit critical limitations that restrict their deployment in roof inspection scenarios. UAV-based systems, while capable of rapid aerial surveys, struggle with close-up inspections and operate at distances that make subtle damage identification difficult (Munasinghe et al. 2024). Their payload capacity and battery life constraints restrict operational time and equipment types, potentially limiting inspection depth and duration (Panigati et al. 2025). Regulatory and safety concerns, especially in urban areas, restrict UAV use due to accident risks or privacy violations, while adverse weather conditions severely affect performance and reliability (Bown and Miller 2018). Ground-based alternatives present their own challenges: wheeled robots lack stability on inclined surfaces exceeding 15° and cannot navigate roof transitions; tracked vehicles offer better terrain adaptability but risk damaging delicate roofing materials through their continuous contact surfaces. Most critically, existing ground robots lack integrated systems that simultaneously address slope adaptation, gait stability, and hazard awareness—three essential requirements for safe roof navigation. This gap motivated our development of a hexapod platform that specifically integrates these capabilities through novel control algorithms rather than relying on conventional locomotion strategies.

Hexapod Robot Platform for Uneven Terrain Tasks

Hexapod robots, with their six-legged design, offer distinct advantages for uneven terrain navigation (Nair 1993), making them particularly suitable for roof inspection. The six-leg configuration provides inherent stability for traversing irregular surfaces, crucial when operating on inclined or unstable roof structures (Tedeschi and Carbone 2015). Their distributed leg structure allows carrying relatively heavy payloads, beneficial for accommodating various sensors and inspection equipment (Bjelonic et al. 2017). Unlike tracked vehicles, hexapods minimize roof surface impact by carefully placing feet, reducing additional damage risk during inspection (Tedeschi and Carbone 2014; Zhao and Jebelli 2025b). The six-leg design provides redundancy, maintaining mobility even if legs malfunction, enhancing operational reliability (Liu et al. 2020). Effective locomotion on inclined surfaces presents unique challenges requiring sophisticated gait strategies and stability control. The tripod gait, where three legs move simultaneously while others provide support, is the most common and fastest but can be compromised on steep inclines (Duan et al. 2009). The wave gait, moving one leg at a time, ensures maximum stability for very steep or slippery surfaces (Xin et al. 2015). Recent research focuses on adaptive gait strategies that dynamically adjust based on inclination and surface conditions (Qiu et al. 2023). Maintaining the robot's center of gravity (CoG) within the support polygon is crucial, involving body orientation modification and individual leg extension adjustment (Gong et al. 2018). Proper force distribution among supporting legs is essential, implementing force sensors to measure ground reaction forces (Zhao et al. 2018) and algorithms to optimize force distribution for enhanced stability and traction (Li et al. 2014). Preventing slipping requires adapting to varying friction conditions, including friction coefficient estimation and modifying stride length, foot placement, and gait patterns for secure footing (Kim et al. 2020). Integrating inclinometers or inertial measurement units (IMUs) enables real-time surface inclination measurement (Zhang et al. 2014) and dynamic gait and posture adaptation (Liu et al. 2019).

Control Systems for Slope Adaption and Collision Avoidance

Fuzzy control systems have emerged as powerful tools for managing complex, nonlinear hexapod dynamics, particularly in maintaining stability on uneven or inclined surfaces. These systems adjust control parameters in real time based on the state of the robot and environmental conditions, effectively handling uncertainties and surface condition variations during roof inspections (Kayacan et al. 2013). Fuzzy logic controllers dynamically modify gait parameters such as step length, lifting height, and cycle time based on sensory feedback, allowing smoother transitions between different surface inclinations and textures (Wang et al. 2017). They continuously optimize stability margins by adjusting body posture and leg configurations, crucial for maintaining balance on steep or slippery surfaces (Kim et al. 2019), while optimizing energy consumption by adapting movements to terrain, essential for prolonged inspection missions (Žák et al. 2023).

Ensuring robotic system safety during roof inspection requires robust edge detection and collision avoidance mechanisms. Combining data from multiple sensor types—LiDAR, RGB-D cameras, and ultrasonic sensors—provides more reliable edge detection (Choi et al. 2013). Advanced algorithms process fused data to create accurate three-dimensional (3D) environmental maps (Yin et al. 2020), enhancing spatial awareness. Deep learning models, such as convolutional neural networks, can be trained to identify roof edges and obstacles from camera images in real time, adapting to various roof types and lighting conditions (Chen et al. 2018). Implementing predictive algorithms that anticipate potential collisions or edge approaches allows proactive avoidance maneuvers (Chai et al. 2017), integrating with gait control to seamlessly adjust locomotion when approaching edges or obstacles. Redundant safety systems, including mechanical stops and emergency shutdown procedures, are crucial for preventing accidents (Sinkar et al. 2018), while incorporating human oversight through remote monitoring provides additional safety layers (You et al. 2024). By integrating these advanced control and safety strategies, hexapod robots can be effectively deployed for roof inspection tasks, navigating complex and potentially hazardous environments, while minimizing risks.

Methodology

This section introduces a comprehensive design of a Slope-adaptive and Hazard-aware (SaHa) hexapod robot system developed for autonomous roof inspection (Fig. 1). This module-enhanced system is built upon an 18 degrees-of-freedom (DoF) six-legged walking robot platform, integrating three key innovations to ensure safe and efficient operation. The base platform incorporates essential hardware components for computational processing, posture sensing, and environmental awareness. The SaHa system's architecture comprises three fundamental modules: a stability-enhanced gait system utilizing Bézier curve trajectories, a slope-adaptive posture control mechanism based on fuzzy logic, and an edge detection system employing infrared sensors. The gait system optimizes foot trajectories to maintain stability on inclined surfaces, while the posture control mechanism continuously adjusts the robot's orientation based on real-time slope measurements. The edge detection system provides comprehensive hazard awareness through strategic sensor placement, enabling safe navigation near roof boundaries. The system operates on the Jetson NANO's quad-core ARM processor, with the gait generator running at 50 Hz, fuzzy controller at 100 Hz, and edge detection at 25 Hz. Total computational load averages 65% CPU utilization during normal operation, increasing to 85% during hazard recovery maneuvers. Module coordination is achieved through a priority-based scheduler by which edge detection commands override gait planning, while posture control continuously adjusts motor commands. During our 20-min continuous operation tests, the system consumed approximately 45 W, enabling 1-h missions with the standard battery pack.

Hexapod Robot Model and Kinematics

Robot Model

This study used the hexapod robot platform based on the model Muto (Yahboom 2024), an 18 DoF six-legged walking robot [Fig. 2(a)]. The robot is equipped with a Jetson NANO board (NVIDIA 2019) for on-board processing. Locomotion is achieved through 18 high-torque servo motors. For precise posture measurement and adjustment, the robot incorporates an IMU, capable of measuring acceleration and gyroscope data. Additionally, the authors strategically placed eight infrared sensors on the robot body for environmental awareness. The hexapod robot has a total weight of approximately 3.5 kg. In its default standing position, it measures 465 mm in length (forward direction) and 514 mm in width (lateral direction). Each of the six legs is composed of three segments: coxa (50 mm), femur (75 mm), and tibia (140 mm).

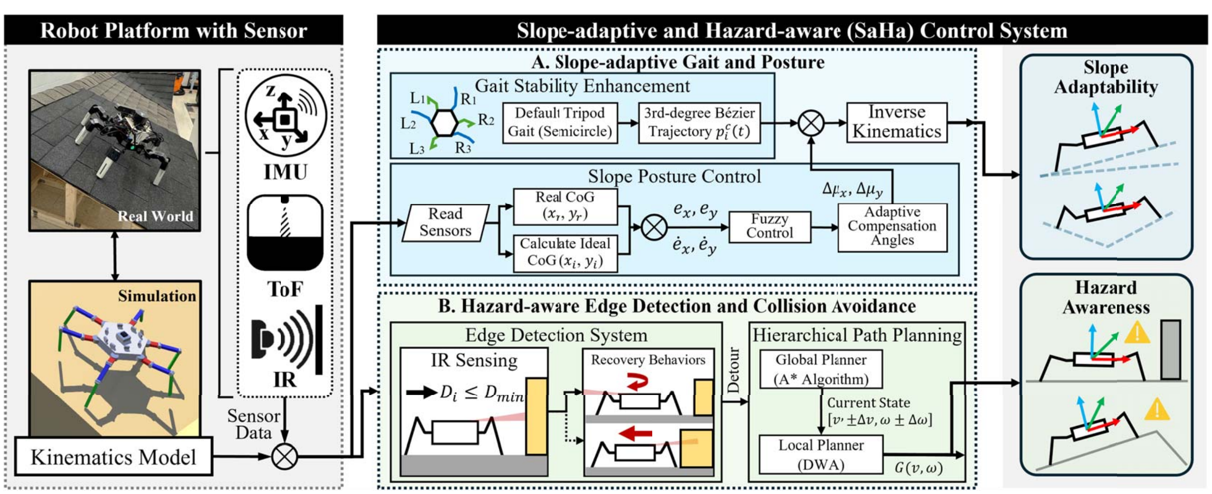
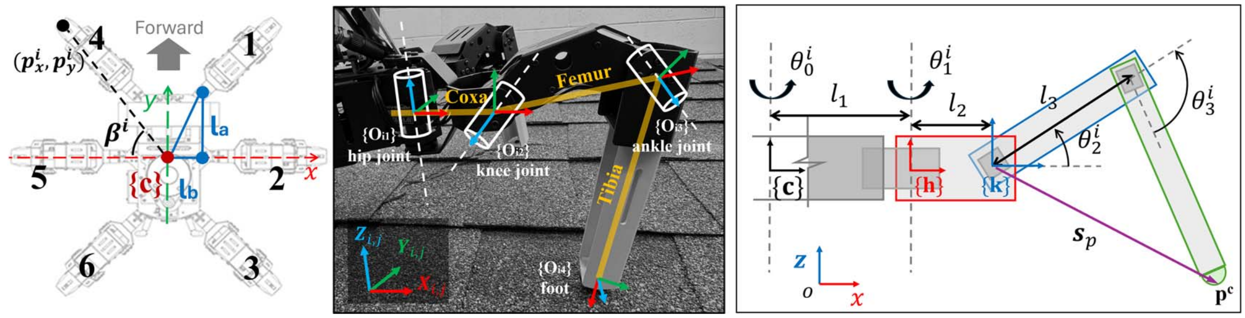


Fig. 1. Module-enhanced hexapod robot control system overview.



(a) (b) (c)

Fig. 2. Hexapod robot kinematic model and limb diagrams: (a) hexapod robot limb setting; (b) coordinate frames; and (c) joint configuration.

The robot's coordinate frame [Fig. 2(b)] is defined as follows: **{c}** is the mass center coordinate frame of the hexapod robot determined by the right-hand rule, with X_c along the body transverse direction, Y_c along the forward direction, and Z_c perpendicular to the ground. $\{O_{ij}\}$ is the coordinate frame of the jth joint of the ith leg of the robot, $i \in \{1,2,3,4,5,6\}$, $j \in \{1,2,3,4\}$. Note when $j = 4$, $\{\mathbf{O_{i4}}\}$ presents the coordinate frame of the foot end. Axis $\boldsymbol{Z_{i,j}}$ is the rotation axis of the joint, the $\boldsymbol{X_{i,j}}$ is chosen along the common normal from axes $\boldsymbol{Z_{i,j}}$ and $\boldsymbol{Z_{i,j-1}}$, and the $\boldsymbol{Y_{i,j}}$ is determined by the right-hand rule. L1, L2, and L3, respectively, represent the length of coxa (C_i), femur (F_i), and tibia (T_i), $i \in \{1,2,3,4,5,6\}$. The joints connecting the leg segments are named as follows: the **hip** joint (angle $\boldsymbol{\theta_1}$) connects the body to the coxa, the **knee** joint (angle $\boldsymbol{\theta_2}$) connects the coxa to the femur, and the **ankle** joint (angle $\boldsymbol{\theta_3}$) connects the femur to the tibia. The parameter $\mathbf{p_i}$ represents the hexapod's feet, which are linked to the tibia ($\mathbf{T_i}$) [Figs. 2(b and c)].

Forward and Inverse Kinematics

The forward kinematics of the 18-DOF hexapod robot is based on the Denavit–Hartenberg (DH) convention. This approach allows for a systematic representation of the spatial relationships between adjacent link coordinate frames, facilitating the calculation of the end-effector position (in this case, the foot position) given the joint angles. The transformation matrix between two adjacent leg coordinate systems is expressed in the DH convention as follows:

$$\mathbf{M}_j^{j-1} = \begin{bmatrix} c_{\theta_j} & -s_{\theta_j} c_{\alpha_j} & s_{\theta_j} c_{\alpha_j} & \alpha_j c_{\theta_j} \\ s_{\theta_j} & c_{\theta_j} s_{\alpha_j} & -c_{\theta_j} s_{\alpha_j} & \alpha_j s_{\theta_j} \\ 0 & s_{\alpha_j} & c_{\alpha_j} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1}$$

where c_{θ_j}, $s_{\theta_j} = \cos(\theta_j)$ and $\sin(\theta_j)$, respectively; and α, a, θ, and d = DH parameters with the particular values listed in Table 1. In Table 1, $\boldsymbol{\alpha}$ is the angle about common normal, from the old z-axis to the new z-axis; $\boldsymbol{a}$ is the length of the common normal; $\boldsymbol{\theta}$ is the angle about the previous z from the old x to the new x; $\boldsymbol{d}$ is the offset along the previous z to the common normal. Let $\mathbf{p^c}$ be the foot-tip position of the leg in the coordinate system relative to the mass center and $\mathbf{p^4}$ be the foot-tip position in the foot-tip coordinate system. Then, the mapping between the body coordinate system and the foot-tip coordinate system is given by the following kinematic chain:

$$\mathbf{p}_i^c = \mathbf{T}_i^c \mathbf{M}_2^1 \mathbf{M}_3^2 \mathbf{M}_4^3 \mathbf{p}_\mathrm{i}^4 \tag{2}$$

where $\mathbf{T}^\mathrm{i}$ = transformation matrix between the body center frame **{c}** and the coxa coordinate frame of the ith leg, given as a rigid body transformation:

$$\mathbf{T}^i = \begin{bmatrix} c_{\beta^i} & -s_{\beta^i} & 0 & p_x^i \\ s_{\beta^i} & -c_{\beta^i} & 0 & p_y^i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}$$

where β^i, p_x^i, and p_y^i, = body parameters of the ith link listed in Fig. 2(a) and Table 2.

Table 1. Denavit–Hartenberg parameters between {c} and the Hexapod's foot $\mathbf{p_i}$

Transformation	αi	a_i	θ_i	d_i
$\{c\} \rightarrow \{O_{i1}\}$	0	l_1	θ_{i1}	0
$\{O_{i1}\} \rightarrow \{O_{i2}\}$	π/2	l_2	θ_{i2}	0
$\{O_{i2}\} \rightarrow \{O_{i3}\}$	0	l_3	θ_{i3}	0
$\{O_{i3}\} \rightarrow \{O_{i4}\}$	0	l_4	θ_{i4}	0

The inverse kinematics for the hexapod robot is simplified by its joint configuration [Fig. 2(c)], allowing for an analytical solution. Given the desired foot-tip coordinates in the global frame, the inverse kinematics calculates the required joint angles for each leg. For a given foot-tip position $\mathbf{p^c} = [p_1^c, p_2^c, p_3^c]$ in the global coordinates, the hip joint angle θ_0^i of the ith is obtained as follows:

$$\theta_0^i = \begin{cases} 0, & i = 2 \\ \arctan\left(\frac{l_a}{l_b}\right), & i \in \{1, 3, 4, 6\} \\ \pi, & i = 5 \end{cases} \tag{4}$$

where l_a and l_b are illustrated in Fig. 2(a). The angle θ_1^i is calculated with respect to the fixed reference frame $\{\mathbf{h}\}$ located at the **hip** joint $\theta_1^i = \arctan(x_p^h/y_p^h)$, and $\mathbf{p}_i^h = \{x_p^h, y_p^h, z_p^h\}^T$ is the position of $\mathbf{p}_i$ relative to $\{\mathbf{h}\}$, obtained by $\mathbf{p}_i^h = (\mathbf{T}_1^0)^{-1}\mathbf{p}_i^c$, in which $\mathbf{T}_1^0$ is the transformation matrix between the center frame $\{\mathbf{c}\}$ and the reference frame $\{\mathbf{h}\}$. To calculate the remaining angles, we need the position of the reference frame $\{\mathbf{k}\}$ at the knee joint, defined by the Denavit–Hartenberg convention as $\mathbf{k}_\mathrm{i}^\mathrm{h} = (\mathbf{T}_1^0)^{-1}\mathbf{T}_2^0$. Considering the vector $\mathbf{s}_\mathrm{p}$ between the knee frame $\{\mathbf{k}\}$ and the foot, the angles

Table 2. Body parameters (units: rad, mm)

1	1	2	3	4	5	6
β^i	π/4	0	7π/4	3π/4	π	3π/2
p_x^i	232.5	257	232.5	−232.5	−257	−232.5
p_y^i	232.5	0	−232.5	232.5	0	−232.5

θ_2^i and θ_3^i are calculated as follows:

$$\theta_2^i = \cos^{-1}\left(\frac{l_3^2 + \|\boldsymbol{s}_p\|^2 - l_4^2}{2l_3\|\boldsymbol{s}_p\|}\right) - \sin^{-1}\left(\frac{z^k - z^p}{\|\boldsymbol{s}_p\|}\right) \tag{5}$$

$$\theta_3^i = \pi + \arccos\left(\frac{l_3^2 + l_4^2 - \|s_p\|^2}{2l_3 l_4}\right) \tag{6}$$

where z^k and z^p = vertical positions of the knee frame $\{k\}$ and the foot, respectively. These equations form the foundation for the robot's motion control and gait design in subsequent sections.

Slope-Adaptive Gait and Posture

Gait Stability Enhancement

The Muto hexapod robot employs a tripod gait characterized by alternating movement patterns in which three legs remain in the stance phase, while the other three are in the swing phase [Fig. 3(a)]. The robot's legs are divided into two sets: Set A comprising legs $R1$, $L2$, and $R3$, and Set B consisting of the remaining legs [Fig. 3(b)]. The tripod gait sequence consists of four steps: maintaining contact with all six feet, moving Set A legs forward while Set B supports, moving Set B legs forward while Set A generates thrust, and lifting Set A legs to return to the initial position [Fig. 3(b)]. This pattern enables continuous locomotion and allows the robot to reverse the sequence for a stationary position. While the tripod gait offers high velocity, it presents stability challenges due to fewer supporting limbs. The swing phase's velocity and acceleration impact torso posture and energy consumption, requiring an algorithm to minimize posture variation. The default trajectory generation algorithm (Algorithm 1) uses a semicircular path for the swing phase and vertical movement for the stance phase but lacks flexibility for varying surfaces and may cause abrupt phase transitions.

Algorithm 1. Semicircle Trajectory Generation.

```
FUNCTION semicircle_generator (radius, steps, reverse = FALSE):
    ASSERT steps is divisible by 4
    half_steps = steps / 2
    step_angle = π / half_steps
    step_stride = 2 * radius / half_steps
    result = empty list
      // Stance Phase: vertical movement (only y-axis changes - leg touching the ground)
        FOR i = 0 TO half_steps - 1
        y = radius - i * step_stride
        ADD (0, round(y, 2), 0) TO result
      // Swing Phase: semicircular movement (y and z change)
    movement (y and z change)
      FOR i = 0 TO half_steps - 1
        angle = π - step_angle * i
        y = radius * cos(angle)
        z = radius * sin(angle)
        ADD (0, round(y, 2), round(z, 2)) TO result
      CONVERT result TO deque
      // Rotate the deque so that y starts from 3/4 of the circle instead of 0
      ROTATE result by steps / 4 positions
      IF reverse is TRUE
        REVERSE result
        ROTATE result by 1 position
      RETURN result
```

To achieve smoother transitions between phases and address stability concerns, particularly when navigating inclined surfaces, our SaHa system employs an enhanced gait system utilizing a Bézier curve trajectory (Algorithm 2).

Algorithm 2. Bézier Curve Trajectory Generation.

```
FUNCTION generate_hexapod_gait_trajectory(t, tint, h, s, L0, L1, L2, L3, θ1, θ2, θ3):
 n = 3 // 3rd-degree Bézier curve
   // Calculate the initial foot position (p_i,0)
   p_i0_x = C(θ2+θ3) * (L3*C(θ2+θ3) + L2*Cθ2 + L1) + L0*Cθ0
   p_i0_y = S(θ2+θ3) * (L3*C(θ2+θ3) + L2*Cθ2 + L1) + L0*Sθ0
   p_i0_z = L3*S(θ2+θ3) + L2*Sθ2
   p_i0 = [p_i0_x, p_i0_y, p_i0_z]
   // Define the remaining control points
   p_i1 = [0, s/4, h/3] + p_i0
   p_i2 = [0, 3s/4, h/3] + p_i0
   p_i3 = [0, s, 0] + p_i0
   // Calculate the Bézier trajectory for the swing phase
   IF t >= 0 AND t <= tint THEN
     p_i_t = (0, 0, 0) // Initialize the trajectory point
       FOR m = 1 TO n
         p_i_t = p_i_t + (t_int - t)^(n-m) * t^m * p_i,m
       END FOR
       RETURN p_i_t
   // Calculate the linear trajectory for the stance phase
     ELSE IF t > t_int THEN
       p_i_t = [0, -s/t_int * t, 0] + p_i0
       RETURN p_i_t
     ELSE
       RETURN "Error: Invalid time input"
     END IF
   END FUNCTION
```

The Bézier curve approach has been demonstrated in the literature (Saputra et al. 2016; Șomîtcă et al. 2022) to provide superior stability for incline walking compared with traditional methods. Based on those findings, our SaHa system adopts a third-degree Bézier curve for its stability-enhancing algorithm. This choice is predicated on its superior performance in terms of roll and pitch velocity variation, improved height stability, and reduced computational overhead compared with higher-degree curves. During the swing phase, the feet move to a new foothold position by following the proposed Bézier trajectory for the ith leg, which can be described as follows:

$$p_i^c(t) = \sum_{m=1}^{n} (t_{\text{int}} - t)^{n-m} t^m p_{i,m}^c,\ t \in [0, t_{\text{int}}],\ i \in [1, 6],\ j \in [0, m] \tag{7}$$

where n = number of control points (curve degree); t_{int} = trajectory's time interval; and $\mathbf{p}_{i,m}^c = \{x_p^c, y_p^c, z_p^c\}^T$ = control points of the trajectory. The first control point corresponds to the foot's initial position $\mathbf{p}_{i,0}^c$ and is defined by Eq. (2), which can also be derived as follows:

$$\mathbf{p}_{i,0}^c = \left\{\begin{array}{c} c_{(\theta_0+\theta_1)}(L_3 c_{(\theta_2+\theta_3)} + L_2 c_{\theta_2} + L_1) + L_0 c_{\theta_0} \\ s_{(\theta_0+\theta_1)}(L_3 c_{(\theta_2+\theta_3)} + L_2 c_{\theta_2} + L_1) + L_0 s_{\theta_0} \\ L_3 s_{(\theta_2+\theta_3)} + L_2 s_{\theta_2} \end{array}\right\} \tag{8}$$

The remaining control points are determined as $\mathbf{p}_{i,1}^c = \{0, s/4, h/3\}^{\mathrm{T}} + \mathbf{p}_{i,0}^c$, $\mathbf{p}_{i,2}^c = \{0, 3s/4, h/3\}^{\mathrm{T}} + \mathbf{p}_{i,0}^c$, $\mathbf{p}_{i,3}^c = \{0, s, 0\}^{\mathrm{T}} + \mathbf{p}_{i,0}^c$, where h is the maximum height, and s denotes the stride length.

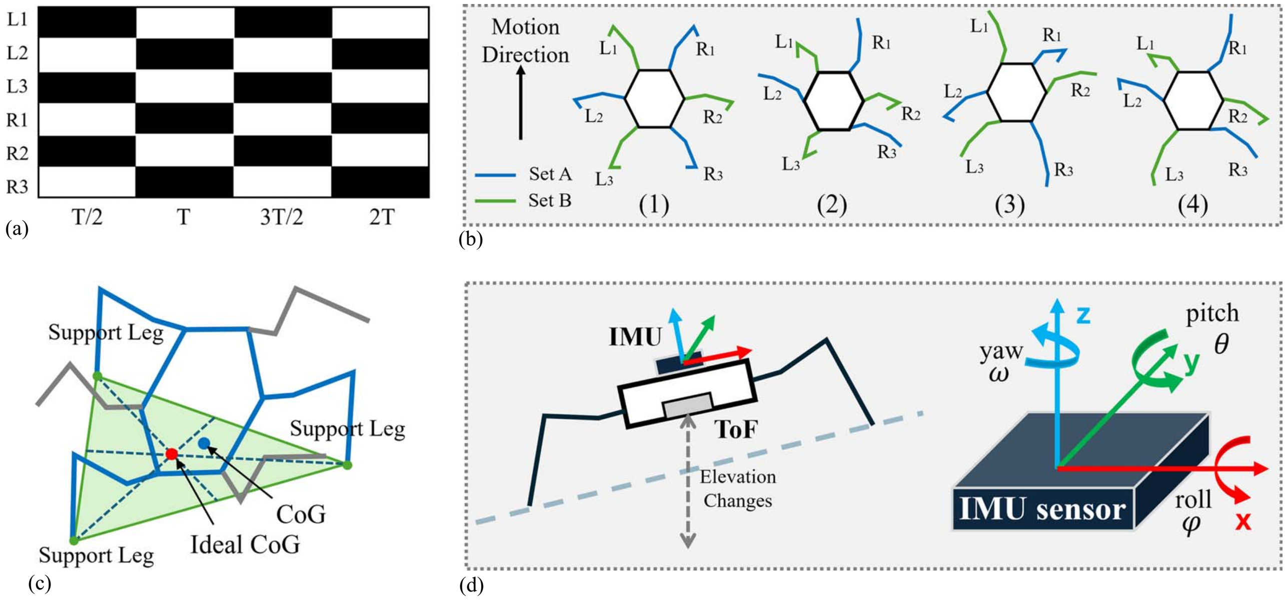


Fig. 3. Robot diagrams: (a) stance and swing phase of a tripod gait; (b) walking sets; (c) diagram of CSP and CoG; and (d) IMU and ToF for body orientation.

During the stance phase, in which the feet maintain ground contact, while propelling the torso forward, the foot trajectory is defined as follows:

$$\mathbf{p}_i^c(\mathrm{t}) = \left\{0, -\frac{s}{t_{\mathrm{int}}}t, 0\right\}^{\mathrm{T}} + \mathbf{p}_{i,0}^c, \; t \in [0, t_{\mathrm{int}}] \tag{9}$$

While these parameters (stride length, foot height, and swing duration) remain fixed during operation, they are carefully balanced to accommodate the range of slopes encountered in roof inspection scenarios (0°–26.6°). This trajectory generation module focuses on providing smooth, stable foot movements, while terrain-specific adaptations are handled by the slope-adaptive posture control system described in the following section.

Posture Control on Slopes

The stability of the hexapod robot on inclined surfaces is crucial for roof inspection. The SaHa system implements slope-adaptive posture control to maintain stability, while navigating roof pitches. The hexapod is statically stable when the vertical projection of its CoG is within the conservative support polygon (CSP) formed by its legs, as shown in Fig. 3(c). For flat surfaces, the D-H convention and forward kinematics calculate support point and CoG projection positions. However, this approach is insufficient on slopes. The robot's reference frame **{c}** is no longer aligned with the global reference frame **{s}**, necessitating a more complex calculation. To address this, a rotation matrix is introduced to transform positions from the robot's reference frame **{c}** to the space reference frame **{s}**:

$$\mathbf{R_{sc}} = \begin{bmatrix} c_\omega c_\theta & c_\omega s_\theta s_\varphi - s_\omega c_\varphi & c_\omega s_\theta c_\varphi + s_\omega s_\varphi & 0 \\ s_\omega c_\theta & s_\omega s_\theta s_\varphi + c_\omega c_\varphi & s_\omega s_\theta c_\varphi - c_\omega s_\varphi & 0 \\ -s_\theta & c_\theta s_\varphi & c_\theta c_\varphi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{10}$$

where the rotation angles φ around the X-axis, θ around the Y-axis, and ω around the Z-axis can be measured by the IMU sensor [Fig. 3(d)], which is settled at the robot's center of mass.

By applying this rotation matrix in conjunction with the D-H convention, the positions of support points in the global reference frame can be calculated: $\begin{bmatrix}\hat{x}_p & \hat{y}_p & \hat{z}_p & 1\end{bmatrix}^T = \mathbf{R}_{sc}\begin{bmatrix}x_p & y_p & z_p & 1\end{bmatrix}^T$. Then, when the robot is walking on a slope, the vertical projection of the CoG and the center of the CSP are then obtained by applying the transformation matrices $\mathbf{T}_i^c$, $\mathbf{M}_2^1$, $\mathbf{M}_3^2$, $\mathbf{M}_4^3$ and the rotation matrix $\mathbf{R_{sc}}$. The detailed calculation can be referred to previous literature (Kayacan et al. 2013; Wang et al. 2017). To maintain stability on inclined surfaces, the SaHa system employs a fuzzy control system that adjusts the angles of the supporting leg motors to keep the vertical projection of the CoG (real CoG) close to the center of the CSP (ideal CoG). The system defines errors e_x and e_y based on the difference between the ideal CoG position (x_i, y_i) and the real CoG projection (x_r, y_r).

A fuzzy proportional-derivative (PD) controller is applied to supporting leg motors. The controller takes as inputs the errors e_x, e_y, and their derivatives $\dot{e}_x$, $\dot{e}_y$. The outputs are compensation angles $\Delta\mu_x$ and $\Delta\mu_y$, which adjust the positions of the supporting legs to correct the CoG projection: $\Delta\mu_x = K_{p,x}e_x + K_{d,x}\dot{e}_x$, $\Delta\mu_y = K_{p,y}e_y + K_{d,y}\dot{e}_y$, where $K_{p,x}$ and $K_{p,y}$ are the proportional gains, $K_{d,x}$ and $K_{d,y}$ are the derivative gains, all of which are dynamically adjusted by the fuzzy PD controller (Table 3). The fuzzy control system achieves dynamic adjustment of proportional and derivative gains based on the magnitude of the errors and their derivatives (Wang et al. 2017). This approach allows the system to respond appropriately to varying conditions. For instance, when errors are large, smaller gains are applied to prevent the robot from overturning. Conversely, when errors are small, larger gains are used for fine-tuning motor adjustments, enhancing stability [Table 4, Figs. 4(a and b)]. The global reference frame rotates

Table 3. Consequent membership functions

Parameter	Value			
	↑	↑↑	↑↑↑	↑↑↑↑
$K_{p,x}$	0.05	0.1	0.15	0.2
$K_{p,y}$	0.2	0.3	0.4	0.5
$K_{d,x}$	0.05	0.07	0.09	0.15
$K_{d,y}$	0.05	0.07	0.09	0.15

Table 4. Fuzzy rule table for the proportional and derivative gains consequent membership functions

$K_{p,x}$ $K_{p,y}$ $K_{d,y}$ $K_{d,x}$	e_y: $--$	$-$	0	$+$	$++$
$\dot{e}_y$					
$--$	↑	↑↑	↑↑↑	↑↑	↑
$-$	↑↑	↑↑↑	↑↑↑↑	↑↑↑	↑↑
0	↑↑↑	↑↑↑↑	↑↑↑↑	↑↑↑↑	↑↑↑
$+$	↑↑	↑↑↑	↑↑↑↑	↑↑↑	↑↑
$++$	↑	↑↑	↑↑↑	↑↑	↑

with the robot's frame around the Z-axis, ensuring effective compensation angles for stable walking on inclines. The fuzzy PD controller dynamically adjusts to real-time errors, outperforming traditional linear PD controllers in challenging environments. It maintains stability through precise motor adjustments, allowing the hexapod robot to walk stably on inclines without detailed dynamic modeling. The slope-adaptive posture control system's effectiveness is shown through experiments on various roof pitches.

Hazard-Aware Edge Detection and Collision Avoidance

Slope adaptability is vital for stable locomotion on inclines, but detecting and avoiding hazards is equally important during roof inspections. Combining slope adaptation with hazard awareness creates a safety framework for reliable autonomous operation in complex roofing environments. This section introduces a hazard-aware system that integrates strategic sensor placement, hierarchical path planning, and real-time obstacle avoidance to enhance slope-adaptive features.

Sensor-Based Edge Detection System

The edge detection system employs eight infrared (IR) sensors strategically positioned around the robot's body [Fig. 5(a)]. This sensor configuration was chosen for its optimal balance of weight, power consumption, and detection reliability. Each sensor features a detection range of 10–50 cm with an approximate beam angle of 30°. The sensor arrangement includes two forward-facing sensors angled at $\pm 15°$ from the robot's longitudinal axis, two rear-facing sensors with similar angular displacement, and four side-mounted sensors (two per side) oriented at $\pm 45°$ from the lateral axis.

For reliable edge detection, our system processes sensor readings through a two-stage filtering approach. First, a moving average filter with a window size of five samples reduces measurement noise. Subsequently, a threshold-based detection algorithm identifies potential edges when the filtered distance measurement exceeds 80% of the sensor's maximum range [$D_{\min} = 30$ cm, as shown in Fig. 5(b)]. To ensure individual leg safety, the system incorporates proximity constraints directly into footstep planning. When sensors detect an edge ($D_i \leq D_{\min}$), the system establishes a virtual no-step zone extending 15 cm from the detected edge boundary based on the sensor's angular position. During gait execution, the inverse kinematics solver checks each planned foot position against these constraints before commanding leg movements. If a planned footstep violates these constraints, the system dynamically adjusts the foot trajectory by reducing stride length ["s" in Eq. (9)] by up to 50%, while maintaining the Bézier curve shape.

The system implements a hierarchical set of recovery behaviors [Fig. 5(b)] when edge detection indicates potential hazards. These behaviors are triggered based on the detection distance D_i relative to $D_{\min}$. When $D_i \leq D_{\min}$ (best timing for detection), the robot initiates rotation recovery, performing an in-place rotation to scan for alternative paths. If edges are detected too late ($D_i << D_{\min}$) or rotation recovery proves insufficient, the system initiates reverse walking, backing along its previous path until sensor readings indicate a safe forward trajectory. As a final safety measure, an emergency stop is triggered when safety constraints cannot be maintained.

Hierarchical Path Planning and Hazard Avoidance

The hazard awareness system implements a hierarchical planning architecture that combines global path planning with local trajectory generation. At the global level, an A* algorithm generates optimal paths between start and goal positions. The modules interact through a shared state machine that prioritizes safety: when edge detection triggers (highest priority), it immediately suspends current gait cycles and initiates recovery behaviors, while the posture controller maintains stability. The fuzzy controller continuously adjusts leg positions based on IMU feedback (10 ms latency), with these adjustments integrated into the next gait cycle computation (20 ms update rate). This coordinated approach ensures smooth transitions between normal locomotion and hazard avoidance while maintaining real-time responsiveness.

The path cost function $f(n)$ is defined as $f(n) = g(n) + h(n)$, where $g(n)$ represents the actual cost from the start node to the current node n, incorporating both distance and elevation changes, and $h(n)$ is a heuristic estimate of the cost from node n to the goal. The heuristic function is modified to account for the robot's kinematic constraints: $h(n) = w_1 D(n, goal) + w_2 \Delta H(n, goal)$, where $D(n, \text{goal})$ is the Euclidean distance to the goal, $\Delta H(n, \text{goal})$ represents the elevation difference, and are weighting factors optimized for roof navigation. At the local level, the dynamic window approach (DWA) (Fox et al. 1997) planner generates and evaluates trajectories within the robot's achievable velocity space (v, ω). The algorithm samples trajectories within a dynamic window [$v \pm \Delta v$, $\omega \pm \Delta\omega$] based on the robot's current state and kinematic constraints.

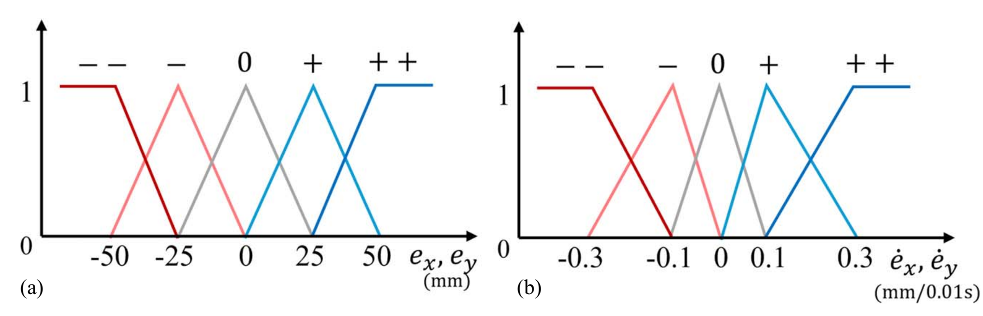


Fig. 4. Premise membership function for the fuzzy control: (a) errors; and (b) derivatives of errors.

ASCE OPEN: Multidiscip. J. Civ. Eng., 2025, 3(1): 04025011

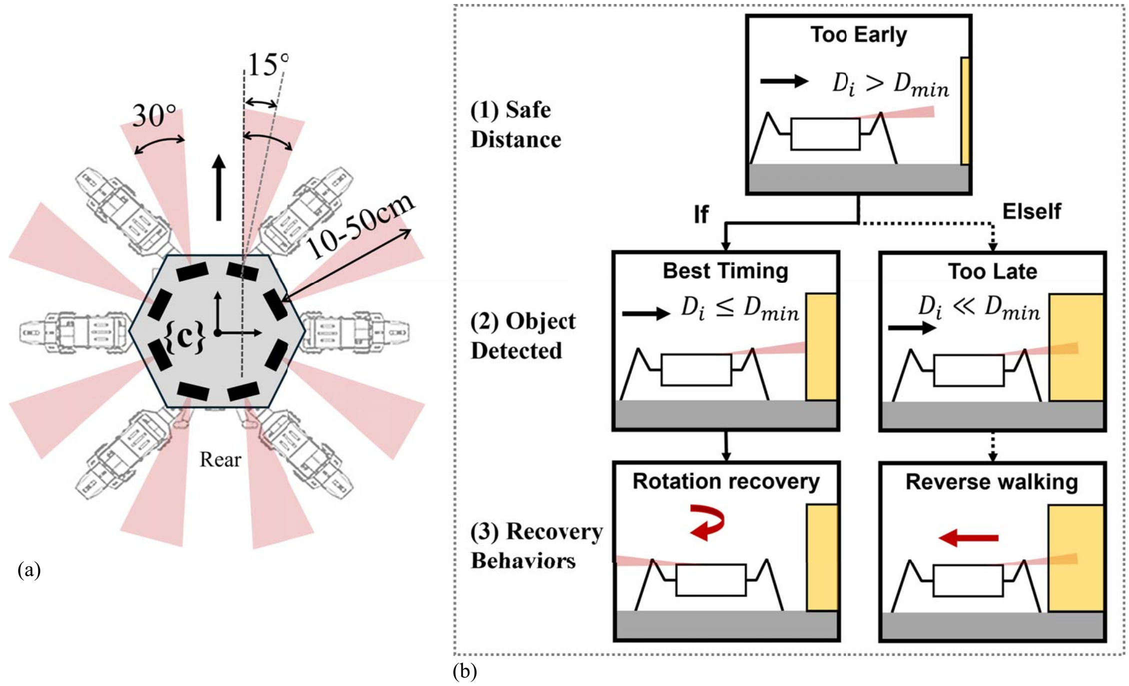


Fig. 5. (a) Eight strategically placed infrared sensors and detection range; and (b) three sequential of the recovery behavior.

Each trajectory is evaluated using a weighted objective function:

$$G(v, \omega) = \alpha_1 \text{ heading}(v, \omega) + \alpha_2 \text{ dist}(v, \omega) + \alpha_3 \text{ vel}(v, \omega) + \alpha_4 \text{ clearance}(v, \omega) \tag{11}$$

The components of this function ensure safe and efficient navigation through heading alignment, goal progress, velocity optimization, and clearance maintenance. The weighting factors are specifically tuned for roof navigation scenarios.

The integration of edge detection with path planning uses a safety framework. When an edge is detected, the safety supervisor updates the cost map and triggers replanning. A safety layer monitors trajectory execution and manages transitions between operation states based on sensor inputs. This integration allows the hexapod robot to navigate complex roof environments, while efficiently progressing toward inspection goals and adapting to various configurations.

Experiments and Validation

The experimental validation of the proposed hexapod robotic system followed a systematic two-phase approach to ensure comprehensive evaluation of its performance and safety features. Phase I consisted of extensive virtual testing conducted in simulated environments utilizing a high-fidelity 3D robot model. The virtual testing environment enabled rapid iteration and refinement of control parameters, while eliminating the risk of hardware damage during the development process. Following the successful completion of virtual testing and parameter optimization, the study progressed to Phase II for preliminary real-world validation.

Phase I: Virtual Experiments

Experimental Setup

The virtual framework evaluated slope adaptability (SA) and hazard awareness (HA) using a Unified Robot Description Format (URDF) 3D model of the hexapod robot developed according to specifications in the previous section, with precise 18-DOF articulated body representation. The simulation incorporated a virtual IMU at the robot's center of mass and a time-of-flight (ToF) laser-ranging sensor, both operating at 100 Hz sampling rate [Fig. 6(c)]. The environment was implemented in Webots with realistic physics parameters (gravity: 9.81 m/s², surface friction: $f = 0.7$) and 0.03-s step size for accurate dynamic behavior representation.

For SA testing, primary indicators included IMU-collected rotation data (roll φ, pitch θ) and ToF-collected elevation data. The stability margin (SM) was calculated as $S_m \equiv \min(D_1, D_2, D_3)$, where D_i represents perpendicular distances from projected CoG to support triangle edges. The SM metric has been extensively validated in previous studies for analyzing hexapod stability (Yang and Kim 1998) and designing fault-tolerant gaits (Wang et al. 2010). For HA testing, metrics included detection distance accuracy (DDA) $= |dM - dA|/dA$, where dM is the measured detection distance and dA is the actual distance to the hazard. Avoidance success rate (ASR) $= (\text{NA}/\text{NTA}) \times 100\%$, where NA represents the number of successful avoidance maneuvers and NTA represents the total number of avoidance scenarios.

SA Testing

The slope adaptability testing protocol was designed to evaluate the effectiveness of the proposed system in enabling stable locomotion across varying roof configurations. Given the complex nature of residential roof structures, the testing was divided into two complementary subtests addressing distinct challenges: continuous slope navigation and structural transition management [Fig. 6(a)].

Subtest 1: Continuous Slopes. The testing evaluated four slope configurations: flat (0°), 2/12 pitch (9.5°), 4/12 pitch (18.4°), and 6/12 pitch (26.6°) on a 5.0 m × 5.0 m platform. The start and goal positions are illustrated in Fig. 6(a). Trials conducted at 0.3 m/s for 10 s compared baseline (default Tripod gait) versus the experimental group (Bézier curve trajectories plus fuzzy control system incorporating the features mentioned in the last section).

ASCE OPEN: Multidiscip. J. Civ. Eng., 2025, 3(1): 04025011

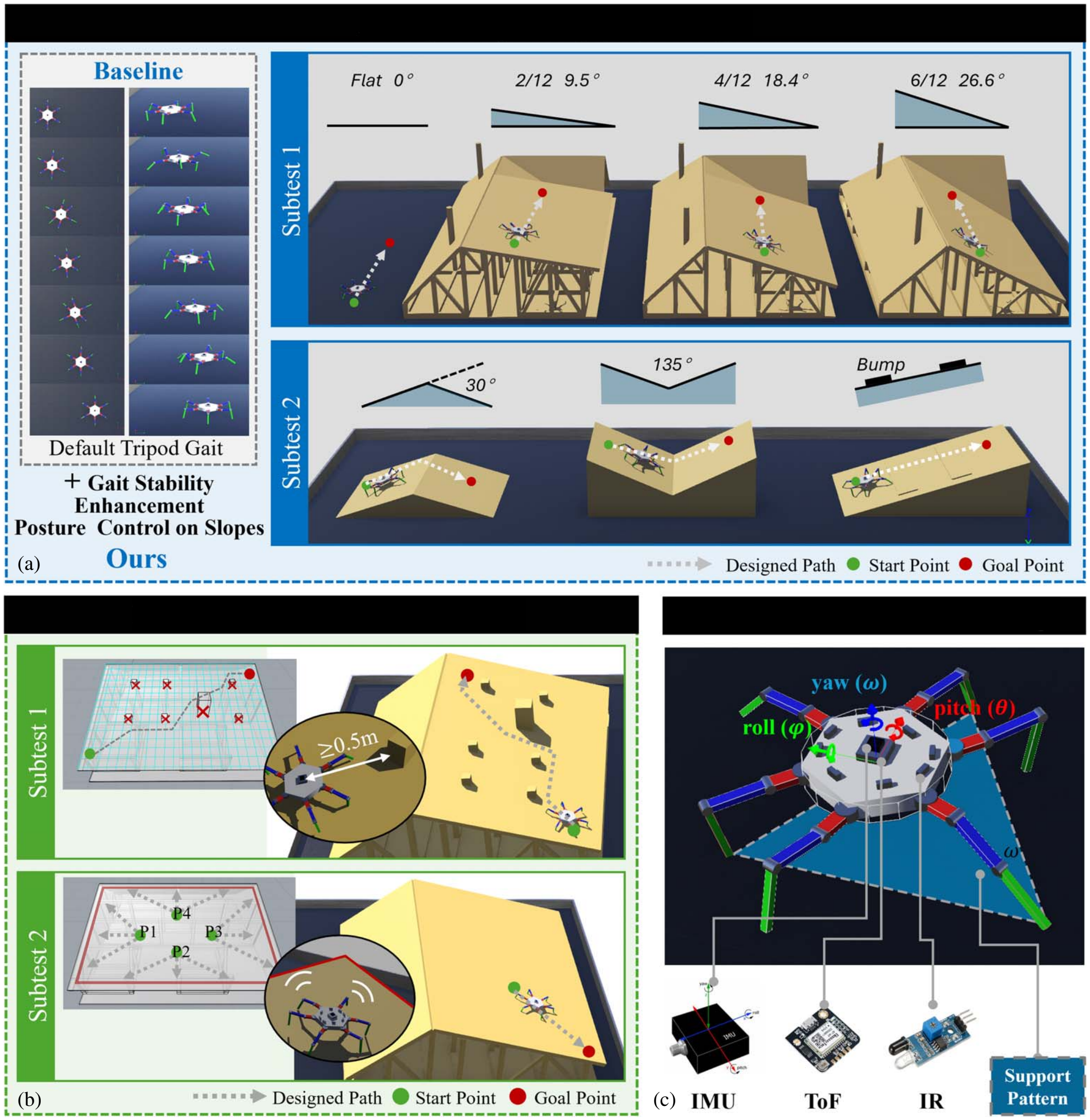


Fig. 6. Slope adaptability (SA) testing: Subtest 1 (continuous slopes), Subtest 2 (slope transitions); Hazard awareness (HA) testing: Subtest 1 (collision avoidance) and Subtest 2 (fall avoidance); the hexapod robot URDF model sensors setting and support pattern diagram.

Subtest 2: Slope Transitions. Testing examined navigation across ridge lines (30° angular transition), valleys (135° angular configuration), and flashing interfaces (10 mm elevation change) as shown in Fig. 6(a). Each transition type was tested twice with both control methods, with success criteria requiring >0.02 m stability margin and $\leq\pm 30°$ maximum body angle deviation.

The combination of these two subtests provided a thorough assessment of the robot's capability to maintain stability and accurate navigation across various surface conditions typically encountered during roof inspections. Data collection throughout both subtests was conducted at a sampling rate of 100 Hz, capturing comprehensive sensor data, including IMU readings, position coordinates, and stability metrics.

HA Testing

The hazard awareness testing protocol was designed to evaluate the effectiveness of the edge detection and collision avoidance system in a controlled simulation environment [Fig. 6(b)]. The testing employed an 8 m × 5 m roof surface at 4/12 pitch with strategically placed obstacles (chimneys: 0.4 m × 0.4 m × 0.5 m; vents: 0.2 m × 0.2 m × 0.15 m) maintaining 1.0 m minimum spacing. The robot model [Fig. 6(c)] featured eight IR sensors configured per previous section (10–50 cm range, 30° beam angle, 25 Hz sampling). Navigation used A^* global planning with 0.1 m resolution grid mapping and DWA local planning at 10 Hz.

Subtest 1: Collision Avoidance. Ten trials from the southwest to the northeast corner (0.5 m edge clearance) evaluated obstacle navigation [Fig. 6(b)]. Success criteria:> 0.5 m obstacle clearance, < 1-minute completion time, and no contact events.

Subtest 2: Fall Avoidance. Testing from four equidistant points (2 m from the roof center) toward nearest edges at 90° and 45° approach angles (0.1 m/s velocity). Success criteria:> 0.25 m minimum stopping distance and $\leq\pm 5°$ maximum body angle deviation during avoidance.

Results and Analysis

SA (Slope Adaptability) Testing Results. *Subtest 1: Continuous Slope Performance.* The experimental results demonstrated progressive performance improvements with increasing surface inclination (Fig. 7–9). For a flat surface (0°), our enhanced system

ASCE OPEN: Multidiscip. J. Civ. Eng., 2025, 3(1): 04025011

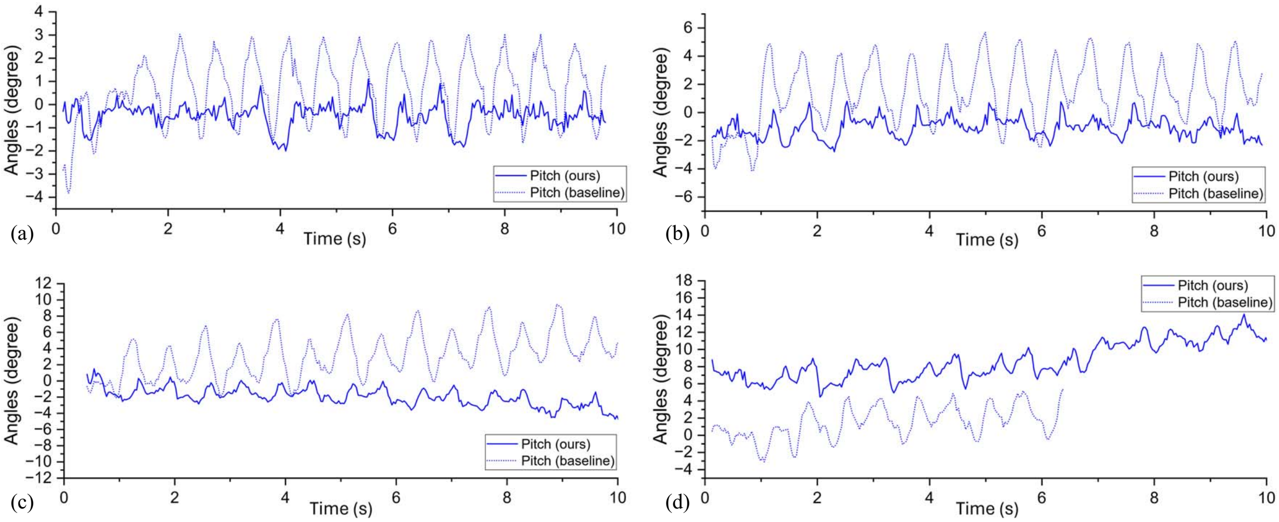


Fig. 7. SA testing Subtest 1 results: pitch angle response across multiple surface inclinations: (a) 0° (flat surface); (b) 9.5° (2:12 slope); (c) 18.4° (4:12 slope); and (d) 26.6° (6:12 slope).

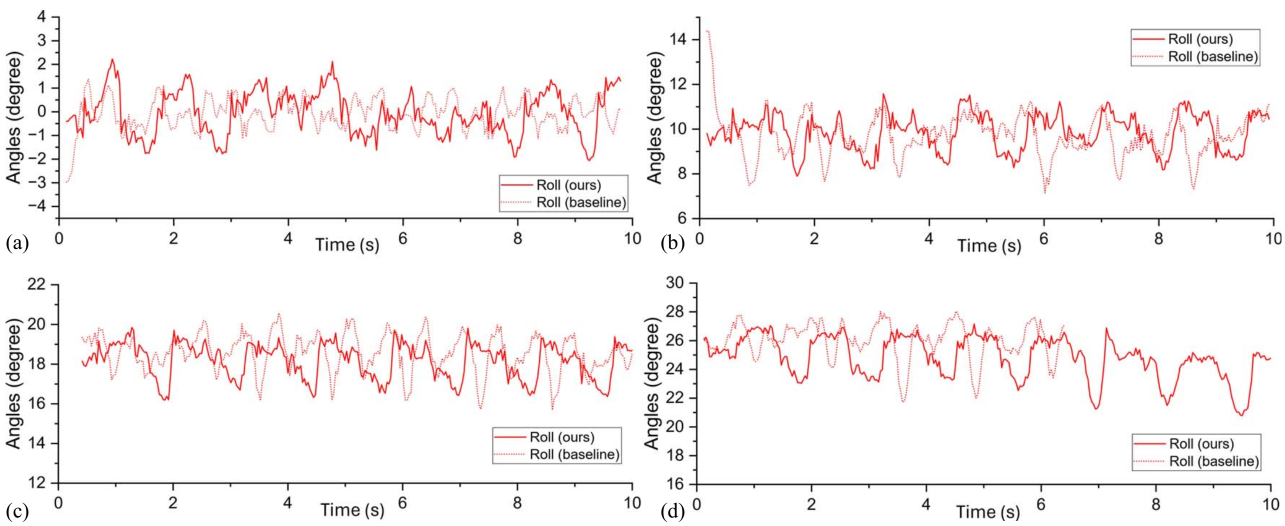


Fig. 8. SA testing Subtest 1 results: roll angle response across multiple surface inclinations: (a) 0° (flat surface); (b) 9.5° (2:12 slope); (c) 18.4° (4:12 slope); and (d) 26.6° (6:12 slope).

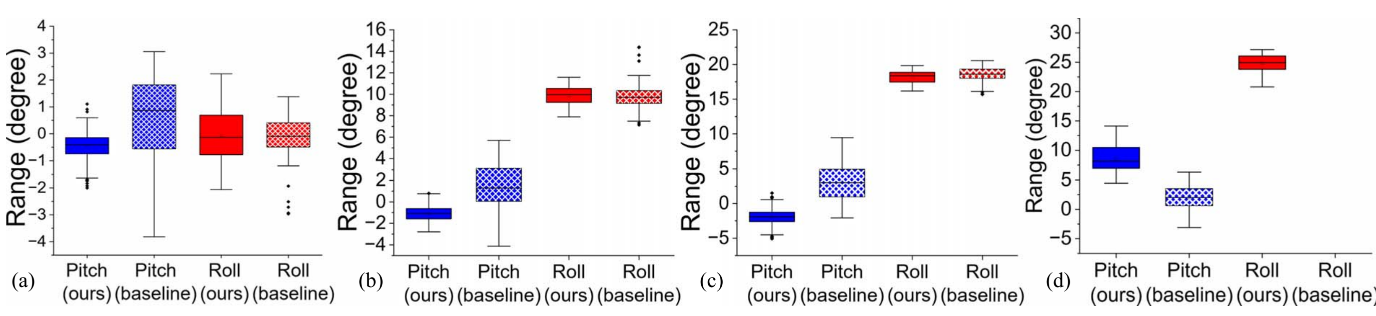


Fig. 9. SA testing Subtest 1 results: pitch and roll distribution analysis across multiple surface inclinations: (a) 0° (flat surface); (b) 9.5° (2:12 slope); (c) 18.4° (4:12 slope); and (d) 26.6° (6:12 slope).

achieved $\pm 0.5°$ pitch control versus the baseline's $\pm 1.8°$ [Fig. 7(a)]. Roll angle distributions showed similar improvement trends [Figs. 8(a) and 9(a)]. For a 2/12 pitch (9.5°), the performance differential became more pronounced with $\pm 2°$ pitch variations compared with the baseline's $\pm 5°$ range [Fig. 7(b)]. For a 4/12 pitch (18.4°), our system maintained $\pm 3°$ pitch variations, while the baseline exhibited $\pm 5°$ deviations [Fig. 7(c)]. For a 6/12 pitch (26.6°), our system maintained $\pm 5°$ pitch control, while

Table 5. SA testing subtest 1 results: mean SM and SD performance results

Slope pitch (angle)	Our method		Baseline tripod method	
	Mean SM (mm)	SD (mm)	Mean SM (mm)	SD (mm)
flat (0°)	**97.5**	**17**	97.4	41.7
2/12 (approximately 9.5°)	**180.9**	**17**	141.8	41.7
4/12 (approximately 18.4°)	**176.2**	**16.5**	132.6	45.3
6/12 (approximately 26.6°)	**171.4**	**17.2**	124.3	46.8

Note: Bolded values highlight the best-performing results.

baseline methods failed to complete trials [Figs. 7(d)]. We also conducted a stability margin analysis; the mean SM and SD results (Table 5) and body center height distribution (Fig. 10) revealed consistent superior performance. Our enhanced system kept stability margins at 97.5 mm (flat), 180.9 mm (9.5°), 176.2 mm (18.4°), and 171.4 mm (26.6°) and outperformed the baseline system's stability margins, which were 97.4 mm (flat), 141.8 mm (9.5°), 132.6 mm (18.4°), and 124.3 mm (26.6°).

Subtest 2: Slope Transition Performance. Structural transition testing (Figs. 11 and 12) demonstrated superior performance across all three transition types. Ridge transitions achieved ± 5° pitch/roll control [Figs. 11(a) and 12(a)] with a 168.7 mm (± 12.3 mm) mean stability margin, while baseline control failed midtraversal. Valley transitions showed −25° to +20° pitch adaptation with ± 15° roll [Figs. 11(b) and 12(b)] and 152.4 mm (± 18.6 mm) stability margin despite the challenging 135° configuration; baseline similarly failed. Flashing transitions maintained a 5°–20° pitch with ± 5° roll control [Figs. 11(c) and 12(c)] and 174.2 mm (± 9.8 mm) stability margin, whereas baseline exhibited excessive deviations and insufficient stability. All enhanced system stability margins exceeded the 20 mm threshold throughout transitions, confirming safe navigation.

Ablation Study. To evaluate the individual contributions of the Bézier gait system and fuzzy posture control, we conducted an ablation study using three configurations on 4/12 pitch (18.4°) slopes: (1) Baseline: default tripod gait without enhancements, (2) Bézier-only: Bézier curve trajectories without fuzzy control, and (3) Fuzzy-only: fuzzy posture control with default gait. Each configuration was tested for 10 trials at 0.3 m/s for 10 s. As given in Table 6, the ablation results demonstrate distinct contributions from each module. The Bézier gait system primarily improved trajectory smoothness, reducing pitch and roll variation by 24% compared with baseline. The fuzzy posture control excelled at stability maintenance, improving the stability margin by 20%, while reducing pitch variation by 16%. However, the synergistic effect of both modules exceeded individual contributions: the combined system achieved 40% reduction in pitch variation and 55% reduction in roll variation compared with baseline, with a 33% improvement in stability margin.

HA Testing Results. *Subtest 1: Collision Avoidance Performance.* Obstacle navigation testing (Tables 7 and 8) achieved robust detection and avoidance capabilities. The system demonstrated 94.3% (± 2.1%) detection distance accuracy with 0.45 m (±0.03 m) average detection range, supporting a 90% path completion rate (9/10 trials). Efficiency metrics showed 21 s (± 7 s) average completion time and 1.32 path efficiency ratio, while maintaining 0.32 m (± 0.08 m) minimum obstacle clearance (Table 7).

Subtest 2: Fall Avoidance Performance. Edge detection testing (Table 8) demonstrated reliable hazard avoidance across approach angles. For 90° approaches, the system achieved a 100% success rate with a 0.28 m (± 0.02 m) detection distance and ± 3.2° body angle deviation. For 45° approaches, performance remained strong with an 87.5% success rate (7/8 trials) and a 0.27 m (± 0.03 m) stopping distance. One system limitation was identified: a failure occurred during 45° uphill approach from Point 2 (NW), indicating vulnerability under certain inclined conditions. The failure resulted from asymmetric leg loading during uphill 45° approaches, causing uneven weight distribution and a partial occlusion of forward-facing IR sensors that delayed edge detection by approximately 0.15 s. Despite this isolated incident, the SaHa system prevented similar failures in all other scenarios through its integrated safety mechanisms: hierarchical path planning with real-time cost map updates, redundant multisensor coverage, and emergency stop protocols that activate at critical stability thresholds.

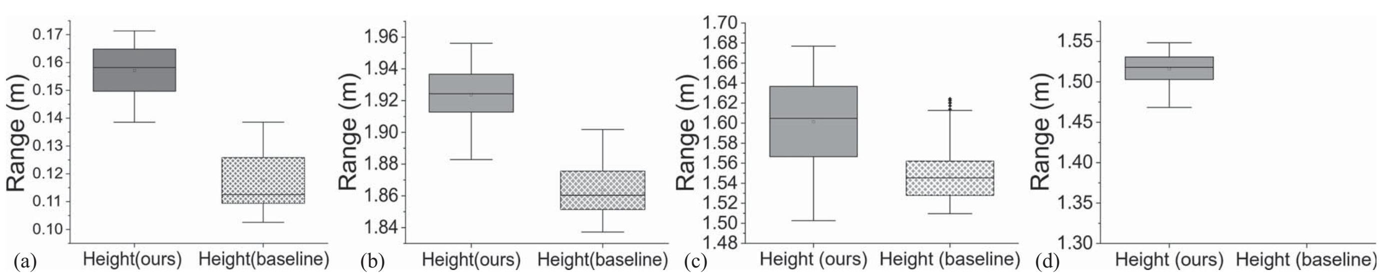


Fig. 10. SA testing Subtest 1 results: body center height distribution analysis across multiple surface inclinations: (a) 0° (flat surface); (b) 9.5° (2:12 slope); (c) 18.4° (4:12 slope); and (d) 26.6° (6:12 slope).

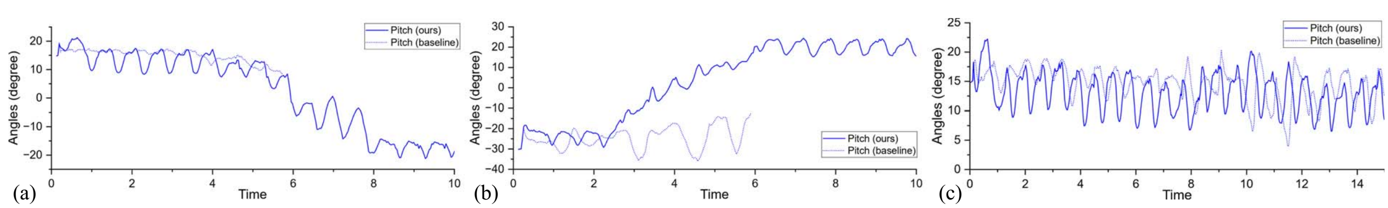


Fig. 11. SA testing Subtest 2 results: pitch angle response compared with the baseline method across multiple slope transition types: (a) ridge line; (b) valley; and (c) flashing interface.

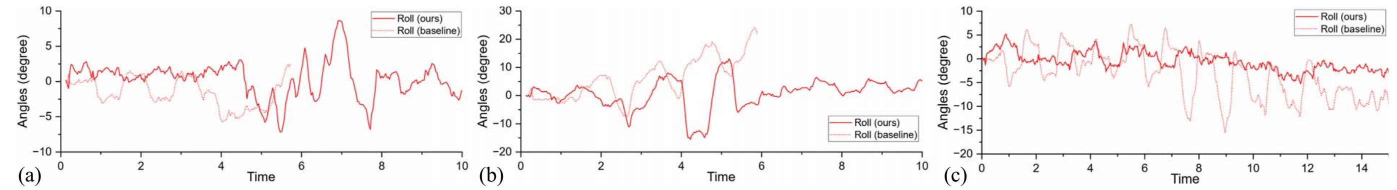


Fig. 12. SA testing Subtest 2 results: roll angle response compared with the baseline method across multiple slope transition types: (a) ridge line; (b) valley; and (c) flashing interface.

Phase II: Preliminary Real-World Validation

Laboratory Setup

The preliminary real-world validation was conducted using a custom-built wooden roof structure measuring 4.0 m (length) × 2.0 m (width) × 0.9 m (height), covered with standard asphalt shingles at 4/12 pitch (18.4°) representing common residential roof slopes (Fig. 13). The physical hexapod robot maintained specifications from the previous section, incorporating the sensor suite with IMU (ICM-20948) mounted at the robot's center of mass (± 0.5° accuracy) and eight IR sensors (Sharp GP2Y0A21YK0F) positioned as mentioned (± 1 cm accuracy at 30 cm range). A high-speed camera system (Sony RX100 VII, 240 fps) positioned perpendicular to the walking path captured detailed motion data for postanalysis.

SA Validation. Slope adaptability validation examined three distinct test paths with both the enhanced system and the baseline method (default tripod gait without fuzzy control). This baseline setting is identical with that used in our virtual experiments, ensuring consistency across testing phases. Success criteria required maintaining >0.02 m stability margin throughout traversal while completing designated paths. All trials conducted at 0.1 m/s with IMU data collected at a 100 Hz sampling rate. The test paths included straight-line walking along a 4/12 pitch slope (1.2 m), ridge line transition (30° angular transition), and valley transition (135° junction). Each path was repeated 10 times with both control methods, yielding 60 total trials (3 paths × 10 repetitions × 2 methods). Reference markers at 0.2 m intervals enabled precise position tracking.

HA Validation. Hazard awareness validation comprised eight edge detection trials from a central starting position. The testing matrix included four cardinal direction approaches (90° angles toward

Table 6. Ablation study evaluating Bézier gait and fuzzy posture control for SA

Configuration	Pitch variation (degrees)	Roll variation (degrees)	Stability margin (mm)
Baseline	± 5.0	± 3.8	132.6 ± 15.2
Bézier-only	± 3.8	± 2.9	145.3 ± 12.1
Fuzzy-only	± 4.2	± 3.2	158.7 ± 13.5
Full system	**± 3.0**	**± 1.7**	**176.2 ± 10.3**

Note: Bolded values highlight the best-performing results.

Table 7. HA virtual testing Subtest 1 results: collision avoidance performance

Performance metric	Value
DDA	94.3% ± 2.1%
Average detection range	0.45 ± 0.03 m
Path completion success rate	90% (9/10)
Average completion time	21 ± 7 s
Path efficiency ratio	1.32
Minimum obstacle clearance	0.32 ± 0.08 m

Table 8. HA virtual testing Subtest 2 results: fall avoidance performance

Starting point	Approach angle	Success	Avg. detection distance (m)	Body angle deviation (degrees)	Stopping distance (m)
Point 1 (NE)	90°	Yes	0.29 ± 0.02	± 3.1	0.27
	45° (downhill)	Yes	0.27 ± 0.02	± 3.8	0.26
	45° (uphill)	Yes	0.26 ± 0.03	± 3.9	0.25
Point 2 (NW)	90°	Yes	0.28 ± 0.02	± 3.2	0.28
	45° (downhill)	Yes	0.27 ± 0.02	± 4.0	0.26
	45° (uphill)	No	—	—	—
Point 3 (SE)	90°	Yes	0.28 ± 0.02	± 3.3	0.27
	45° (west)	Yes	0.26 ± 0.03	± 4.1	0.26
	45° (east)	Yes	0.25 ± 0.03	± 4.0	0.25
Point 4 (SW)	90°	Yes	0.27 ± 0.02	± 3.2	0.28
	45° (west)	Yes	0.26 ± 0.02	± 3.9	0.27
	45° (east)	Yes	0.25 ± 0.03	± 4.1	0.26

Table 9. SA real-world validation results

Path type	Method	Avg. pitch variation (degrees)	Avg. roll variation (degrees)	Avg. body height (m)	Success rate
Straight line	Ours	**± 3.8**	**± 2.9**	0.23 ± 0.01	**9/10**
	Baseline	± 4.2	± 3.1	0.22 ± 0.02	6/10
Ridge line	Ours	**± 3.5**	**± 2.4**	0.22 ± 0.01	**6/10**
	Baseline	± 5.8	± 4.2	0.21 ± 0.03	2/10
Valley	Ours	**± 4.2**	**± 2.8**	0.22 ± 0.02	**7/10**
	Baseline	± 6.5	± 4.8	0.20 ± 0.03	3/10

Note: Bolded values highlight the best-performing results.

Table 10. HA real-world validation results

Approach direction	Approach angle (degrees)	Detection distance (m)	Stopping distance (m)	Body angle deviation (degrees)
North	90	0.26 ± 0.03	0.28 ± 0.02	± 3.2
South	90	0.25 ± 0.04	0.29 ± 0.04	± 3.5
East	90	0.24 ± 0.02	0.27 ± 0.02	± 3.0
West	90	0.24 ± 0.03	0.28 ± 0.02	± 3.2
Northeast	45	0.23 ± 0.04	0.26 ± 0.03	± 4.1
Northwest	45	0.24 ± 0.04	0.25 ± 0.05	± 3.9
Southeast	45	0.21 ± 0.04	0.26 ± 0.03	± 3.7
Southwest	45	0.23 ± 0.04	0.25 ± 0.05	± 3.8

north, south, east, and west) and four diagonal approaches (45° angles toward northeast, northwest, southeast, and southwest). Because of the physical constraints of the test platform (4.0 m × 2.0 m), all trials initiated from a central starting position to ensure adequate walking distance before edge detection. Each direction was tested three times to ensure statistical reliability. All trials conducted at 0.1 m/s until edge detection occurred, with position tracking via onboard encoders and external reference markers at 0.1 m intervals. Detection events were recorded through internal logging and external high-speed video capture for precise correlation.

Results and Analysis

SA Validation Results. Our enhanced system for SA validation showed superior stability across all test paths, with significant improvements in stability metrics compared with the baseline (Table 9). In straight-line traversal, it achieved low pitch and roll variations of $\pm 3.8°$ and $\pm 2.9°$, representing 33% and 39% improvements, respectively. During ridge line navigation, it demonstrated pitch and roll variations of $\pm 3.5°$ and $\pm 2.4°$, translating to a 40% improvement. In valley line scenarios, variations were $\pm 4.2°$ in pitch and $\pm 2.8°$ in roll, a 35% enhancement. The system's success rates were high: straight line (9/10), valley line (7/10), and ridge lines (6/10), while the baseline had low success rates in ridge (2/10) and valley line (3/10). Body height maintenance was consistent across terrains (0.22–0.23 m $\pm$ 0.01–0.02 m).

HA Validation Results. For HA validation (Table 10), our system succeeds across all eight approach directions. The 90° approaches demonstrated higher detection distances (average 0.25 m $\pm$ 0.03 m) than 45° approaches (average 0.23 m $\pm$ 0.04 m), with a systematic difference of approximately 2 cm aligning with the sensor arrangement and approach angle. All approaches maintained stopping distances above the critical 0.28 m safety threshold, with 90° averaging 0.275 m and 45° averaging 0.255 m. Body angle deviations were tighter for 90° approaches (average $\pm 3.2°$) compared with 45° approaches (average $\pm 3.9°$).

Discussion

The SaHa hexapod robot system advances autonomous roof inspection with Bézier curve trajectory generation, fuzzy logic posture control, and hazard detection. Testing showed improvements in stability, adaptability, and safety. The gait system using third-degree Bézier curves achieved 35% reduction in pitch variation and 37% in roll variation during walking. The fuzzy logic controller allowed a navigation of roof pitches up to 6/12 (26.6°), expanding the operational range. The hazard awareness system maintained over 90% detection accuracy, ensuring safe distances from edges and obstacles during inspections.

Comparing Phase I virtual experiments with Phase II preliminary real-world validation revealed notable performance differences providing valuable implementation insights. Virtual experiments on 4/12 pitch roofs showed pitch/roll variations within $\pm 3.0°/\pm 1.7°$, while real-world trials exhibited slightly elevated variations ($\pm 3.8°/\pm 2.9°$) and reduced ridge transition success rates (6/10 versus higher virtual performance). Discrepancies stem from unmodeled environmental factors like shingle variations, sensor noise, and mechanical tolerances. The HA module also showed simulation-reality gaps. Phase I had near-perfect detection (0.28 m $\pm$ 0.02 m), while Phase II saw slight degradation (0.23 m $\pm$ 0.04 m) due to sensor occlusion and ambient interference,

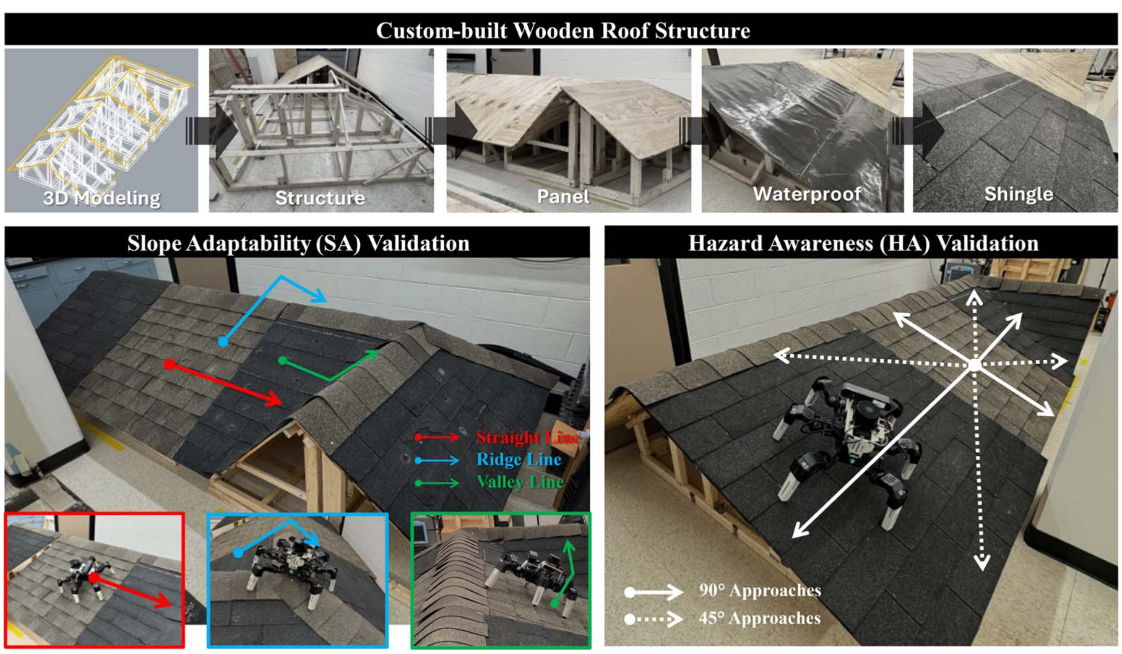


Fig. 13. Custom-built wooden roof structure and two validation method overview.

ASCE OPEN: Multidiscip. J. Civ. Eng., 2025, 3(1): 04025011

particularly with diagonal uphill edge approaches where latency and asymmetric leg loading caused instability—issues not fully anticipated in virtual testing.

While the current locomotion framework demonstrates stable and reliable performance across a range of inclined surfaces (0°–26.6°), one limitation lies in the use of manually defined Bézier trajectory parameters—namely stride length, foot height, and swing duration—which remain fixed regardless of changes in terrain conditions such as slope, friction, or surface irregularities. These parameters were carefully tuned through empirical testing to balance stability and computational efficiency for the specific hardware configuration of our hexapod robot. However, the lack of real-time adaptation in these gait parameters limits the system's responsiveness to more complex, variable, or rapidly changing environments.

Although our slope-adaptive posture control module partially compensates for terrain variability by adjusting body orientation and leg support angles, it does not fully address cases in which dynamic adjustments to foot trajectories could further enhance locomotion performance and resilience. Integrating sensor-based or learning-driven methods to enable real-time modulation of gait trajectories based on environmental feedback represents a promising direction for future research. Also, the controlled experimental conditions excluded adverse weather scenarios (rain, wind) pivotal for practical deployment. Future research should prioritize bridging the simulation-to-reality gap through adaptive models integrating real-time friction estimation and machine learning–driven terrain interaction. Training the fuzzy controller on dynamic surface conditions could enhance responsiveness to unexpected traction changes. Redundant sensing architectures with advanced fusion algorithms could bolster hazard detection robustness. Expanding testing to irregular geometries (domed roofs) and varied materials (metal, tile) would validate system generalizability, while energy efficiency optimization could extend operational endurance for prolonged missions.

Conclusion

This study presents a comprehensive development and evaluation of a slope-adaptive and hazard-aware (SaHa) hexapod robot system for autonomous roof inspection. Through an integration of Bézier curve trajectory generation, fuzzy logic–based posture control, and strategic hazard detection, the system demonstrated significant advancements in robotic stability, adaptability, and safety during roof navigation. The experimental validation through virtual simulation and real-world testing provided strong evidence of system capabilities. The enhanced control system achieved 33%–39% reductions in pitch and roll variations during straight-line walking compared with baseline methods, maintained consistent performance across roof pitches up to 6/12 (26.6°), and achieved detection accuracy rates above 90%, while maintaining safe distances from edges and obstacles. The parallel testing approach revealed valuable insights into practical implementation challenges, with core functionality and safety features successfully validated across both testing phases despite some performance differences between simulated and real-world environments. Several promising research directions emerge from this work. Enhanced simulation models incorporating detailed material properties and environmental factors could bridge virtual-physical performance gaps. The fuzzy control system could be evolved from its current discrete implementation to incorporate continuous membership functions or learning-based adaptation, potentially through reinforcement learning or Adaptive Neuro-Fuzzy Inference System (ANFIS) approaches. Such enhancements would require more powerful computational platforms but could provide superior generalization across diverse terrain conditions.

The successful SaHa system implementation represents a significant step toward addressing roof inspection safety challenges. While further development is needed to address identified limitations, this work provides a robust foundation for future autonomous building inspection systems. The methodologies and insights contribute not only to roof inspection but also to broader autonomous robot development for complex construction and maintenance tasks, with the potential to significantly reduce workplace injuries and improve maintenance efficiency in the construction sector.

Data Availability Statement

All data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgments

This work was supported by the National Science Foundation Award (Grant Nos. ECCS-2525502 and ECCS-2525503, "Collaborative Research: Endowing Aerial Robots with Ground Mobility and Multimodal Perception for Autonomous Roof Inspection"). Any opinions, findings, conclusions, or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

Author Contributions

Xiayu Zhao: Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing—original draft; Yizhi Liu: Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources; Houtan Jebelli: Conceptualization, Methodology, Project administration, Resources, Supervision, Writing—review and editing.

References

Bjelonic, M., T. Homberger, N. Kottege, P. Borges, M. Chli, and P. Beckerle. 2017. "Autonomous navigation of hexapod robots with vision-based controller adaptation." In *Proc., 2017 IEEE Int. Conf. on Robotics and Automation*, 5561–5568. Singapore: IEEE.

BLS (Bureau of Labor Statistics). 2021. *National Census of Fatal Occupational Injuries in 2020*. Washington, DC: Dept. of Labor.

Bock, T. 2015. "The future of construction automation: Technological disruption and the upcoming ubiquity of robotics." *Autom. Constr.* 59: 113–121. https://doi.org/10.1016/j.autcon.2015.07.022.

Bown, M., and K. Miller. 2018. "The use of unmanned aerial vehicles for sloped roof inspections—Considerations and constraints." *J. Facil. Manage. Educ. Res.* 2 (1): 12–18. https://doi.org/10.22361/jfmer/93832.

Chai, X., F. Gao, C. Qi, Y. Pan, Y. Xu, and Y. Zhao. 2017. "Obstacle avoidance for a hexapod robot in unknown environment." *Sci. China Technol. Sci.* 60 (6): 818–831. https://doi.org/10.1007/s11431-016-9017-6.

Chen, Y., W. Li, C. Sakaridis, D. Dai, and L. Van Gool. 2018. "Domain adaptive faster R-CNN for object detection in the wild." In *Proc., IEEE Conf. on Computer Vision and Pattern Recognition*, 3339–3348. Salt Lake City, Utah: Computer Vision Foundation, IEEE.

Choi, C., A. J. B. Trevor, and H. I. Christensen. 2013. “RGB-D edge detection and edge-based registration.” In *Proc., 2013 IEEE/RSJ Int. Conf. on Intelligent Robots and Systems*, 1568–1575. Tokyo, Japan: IEEE

Duan, X., W. Chen, S. Yu, and J. Liu. 2009. “Tripod gaits planning and kinematics analysis of a hexapod robot.” In *Proc., 2009 IEEE Int. Conf. on Control and Automation*, 1850–1855. Christchurch, New Zealand: IEEE.

Fior Markets. 2019. “Global robotics market by robotic type, component, structure, application, region, industry analysis, size, share, growth, trends, and forecast 2018 to 2025—Fior markets.” Accessed October 22, 2024. https://www.fiormarkets.com/report/global-robotics-market-by-robotic-type-industrial-robots-376027.html.

Fox, D., W. Burgard, and S. Thrun. 1997. “The dynamic window approach to collision avoidance.” *IEEE Rob. Autom. Mag.* 4 (1): 23–33. https://doi.org/10.1109/100.580977.

Ge, L., and A. Sadhu. 2025. “Deep learning-enhanced smart ground robotic system for automated structural damage inspection and mapping.” *Autom. Constr.* 170: 105951. https://doi.org/10.1016/J.AUTCON.2024.105951.

Gong, D., P. Wang, S. Zhao, L. Du, and Y. Duan. 2018. “Bionic quadruped robot dynamic gait control strategy based on twenty degrees of freedom.” *IEEE/CAA J. Autom. Sin.* 5 (1): 382–388. https://doi.org/10.1109/JAS.2017.7510790.

Halder, S., and K. Afsari. 2023. “Robots in inspection and monitoring of buildings and infrastructure: A systematic review.” *Appl. Sci.* 13: 2304. https://doi.org/10.3390/app13042304.

He, J., and F. Gao. 2020. “Mechanism, actuation, perception, and control of highly dynamic multilegged robots: A review.” *Chin. J. Mech. Eng.* 33 (1): 1–30. https://doi.org/10.1186/S10033-020-00485-9/FIGURES/17.

Kayacan, E., E. Kayacan, H. Ramon, and W. Saeys. 2013. “Adaptive neuro-fuzzy control of a spherical rolling robot using sliding-mode-control-theory-based online learning algorithm.” *IEEE Trans. Cybern.* 43 (1): 170–179. https://doi.org/10.1109/TSMCB.2012.2202900.

Kim, D., D. Carballo, J. Di Carlo, B. Katz, G. Bledt, B. Lim, and S. Kim. 2020. “Vision aided dynamic exploration of unstructured terrain with a small-scale quadruped robot.” In *Proc., 2020 IEEE Int. Conf. on Robotics and Automation*, 2464–2470. Paris, France: IEEE.

Kim, D., J. Di Carlo, B. Katz, G. Bledt, and S. Kim. 2019. “Highly dynamic quadruped locomotion via whole-body impulse control and model predictive control.” Preprint, Submitted September 14, 2019. http://arxiv.org/abs/1909.06586.

Lattanzi, D., and G. Miller. 2017. “Review of robotic infrastructure inspection systems.” *J. Infrastruct. Syst.* 23 (3): 04017004. https://doi.org/10.1061/(ASCE)IS.1943-555X.0000353.

Li, M., Z. Jiang, P. Wang, L. Sun, and S. Sam Ge. 2014. “Control of a quadruped robot with bionic springy legs in trotting gait.” *J. Bionic Eng.* 11 (2): 188–198. https://doi.org/10.1016/S1672-6529(14)60043-3.

Liang, C. J., T. H. Le, Y. Ham, B. R. K. Mantha, M. H. Cheng, and J. J. Lin. 2024. “Ethics of artificial intelligence and robotics in the architecture, engineering, and construction industry.” *Autom. Constr.* 162: 105369. https://doi.org/10.1016/J.AUTCON.2024.105369.

Liu, C., Z. Li, C. Zhang, Y. Yan, and R. Zhang. 2019. “Gait planning and control for a hexapod robot on uneven terrain based on Markov decision process.” In *Proc., 2019 14th IEEE Conf. on Industrial Electronics and Applications*, 583–586. Xi’an, China: IEEE.

Liu, M., J. Guo, L. Deng, S. Wang, and H. Wang. 2025. “Enhanced vision-based 6-DoF pose estimation for robotic rebar tying.” *Autom. Constr.* 171: 105999. https://doi.org/10.1016/J.AUTCON.2025.105999.

Liu, Y., M. Habibnezhad, and H. Jebelli. 2021. “Brain-computer interface for hands-free teleoperation of construction robots.” *Autom. Constr.* 123: 103523. https://doi.org/10.1016/j.autcon.2020.103523.

Liu, Y., X. Fan, L. Ding, J. Wang, T. Liu, and H. Gao. 2020. “Fault-tolerant tripod gait planning and verification of a hexapod robot.” *Appl. Sci.* 10 (8): 2959. https://doi.org/10.3390/app10082959.

Mirjan, A., F. Gramazio, and M. Kohler. 2019. “Building with flying robots.” *Fabricate* 2014: 266–271. https://doi.org/10.2307/J.CTT1TP3C5W.36.

Miura, H., T. Aridome, and M. Matsuoka. 2020. “Deep learning-based identification of collapsed, non-collapsed and blue tarp-covered buildings from post-disaster aerial images.” *Remote Sens.* 12 (12): 1924. https://doi.org/10.3390/rs12121924.

Munasinghe, I., A. Perera, and R. C. Deo. 2024. “A comprehensive review of UAV-UGV collaboration: Advancements and challenges.” *J. Sensor Actuator Networks* 13 (6): 81. https://doi.org/10.3390/JSAN13060081.

Munawar, H. S., F. Ullah, A. Heravi, M. J. Thaheem, and A. Maqsoom. 2021. “Inspecting buildings using drones and computer vision: A machine learning approach to detect cracks and damages.” *Drones* (6): 5. https://doi.org/10.3390/DRONES6010005.

Nair, S. S. 1993. “Overview of the features of a legged locomotion system for unstructured environments.” *Autom. Constr.* 1 (4): 361–370. https://doi.org/10.1016/0926-5805(93)90017-R.

NVIDIA. 2019. “Jetson Nano.” Accessed November 16, 2024. https://developer.nvidia.com/embedded/jetson-nano.

Ojha, A., Y. Zhang, and H. Jebelli. 2025. “Transforming construction robotics education with virtual reality: Analyzing student presence and self-efficacy.” In *Proc., 42nd Int. Symp. on Automation and Robotics in Construction*. Montreal, Canada: Concordia University.

Panigati, T., M. Zini, D. Striccoli, P. F. Giordano, D. Tonelli, M. P. Limongelli, and D. Zonta. 2025. “Drone-based bridge inspections: Current practices and future directions.” *Autom. Constr.* 173: 106101. https://doi.org/10.1016/J.AUTCON.2025.106101.

Qiu, Z., W. Wei, and X. Liu. 2023. “Adaptive gait generation for hexapod robots based on reinforcement learning and hierarchical framework.” *Actuators* 12 (2): 75. https://doi.org/10.3390/act12020075.

Ren, T., and H. Jebelli. 2025. “Efficient 3D robotic mapping and navigation method in complex construction environments.” *Comput.-Aided Civ. Infrastruct. Eng.* 40 (12): 1580–1605. https://doi.org/10.1111/MICE.13353.

Ridgeline. 2023. “Roofing industry statistics and facts (2023 update).” Accessed October 22, 2024. https://ridgelineconstructionhsv.com/roofing-industry-statistics-and-facts-2023-update/.

Saputra, A. A., N. N. W. Tay, Y. Toda, J. Botzheim, and N. Kubota. 2016. “Bezier curve model for efficient bio-inspired locomotion of low cost four legged robot.” In *Proc., IEEE Int. Conf. on Intelligent Robots and Systems*, 4443–4448. Daejeon, Korea: IEEE.

Sinkar, A., A. Pandey, C. Mehta, K. S. S, A. K. Jayachandran, and I. A. Palani. 2018. “Design and development of wall climbing hexapod robot with SMA actuated suction gripper.” *Procedia Comput. Sci.* 133: 222–229. https://doi.org/10.1016/j.procs.2018.07.027.

Șomîtcă, I. A., S. Brad, V. Florian, and Ș. E. Deaconu. 2022. “Improving path accuracy of mobile robots in uncertain environments by adapted Bézier curves.” *Electronics* 11 (21): 3568. https://doi.org/10.3390/ELECTRONICS11213568.

Sutley, E. J., K. Vazquez, J. H. Kim, T. Dao, B. Johnston, and J. Hunt. 2020. “Performance of manufactured housing during hurricanes Irma and Michael.” *J. Perform. Constr. Facil.* 34 (4): 04020078. https://doi.org/10.1061/(ASCE)CF.1943-5509.0001486.

Tedeschi, F., and G. Carbone. 2014. “Design issues for hexapod walking robots.” *Robotics* 3 (2): 181–206. https://doi.org/10.3390/robotics3020181.

Tedeschi, F., and G. Carbone. 2015. “Hexapod walking robot locomotion.” In *Motion and operation planning of robotic systems: Background and practical approaches*, edited by G. Carbone and F. Gomez-Bravo, 439–468. Cham, Switzerland: Springer.

Wang, R., M. Fu, J. Yu, Y. Yang, and W. Song. 2024. “Risk-inspired aerial active exploration for enhancing autonomous driving of UGV in unknown off-road environments.” In *Proc., IEEE Int Conf Robot Autom*, 14390–14396. Yokohama, Japan: IEEE.

Wang, W.-J., H.-G. Chou, Y.-J. Chen, and R.-C. Lu. 2017. “Fuzzy control strategy for a hexapod robot walking on an incline.” *Int. J. Fuzzy Syst.* 19 (6): 1703–1717. https://doi.org/10.1007/s40815-017-0319-9.

Wang, Z.-Y., X.-L. Ding, and A. Rovetta. 2010. “Analysis of typical locomotion of a symmetric hexapod robot.” *Robotica* 28 (6): 893–907. https://doi.org/10.1017/S0263574709990725.

Xin, G., H. Deng, G. Zhong, and H. Wang. 2015. “Hierarchical kinematic modelling and optimal design of a novel hexapod robot with integrated limb mechanism." *Int. J. Adv. Rob. Syst.* 12 (9): 123. https://doi.org/10.5772/59989.

Yahboom. 2024. "Muto RS robot." Accessed May 26, 2025. http://www.yahboom.net/study/Muto-RS.

Yang, J.-M., and J.-H. Kim. 1998. "Fault-tolerant locomotion of the hexapod robot." *IEEE Trans. Syst. Man Cybern. Part B Cybern.* 28 (1): 109–116. https://doi.org/10.1109/3477.658585.

Yin, H., Y. Wang, X. Ding, L. Tang, S. Huang, and R. Xiong. 2020. "3D LiDAR-based global localization using Siamese neural network." *IEEE Trans. Intell. Transp. Syst.* 21 (4): 1380–1392. https://doi.org/10.1109/TITS.2019.2905046.

You, B., X. Chen, J. Li, L. Ding, and Z. Dong. 2024. *Human-robot collaborative decision method of hexapod robot based on prior knowledge and negotiation strategy*. New York: Knowledge-Based Systems.

Žák, M., J. Rozman, and F. V. Zbořil. 2023. "Energy efficiency of a wheeled bio-inspired hexapod walking robot in sloping terrain." *Robotics* 12 (2): 42. https://doi.org/10.3390/robotics12020042.

Zhang, J., F. Gao, X. Han, X. Chen, and X. Han. 2014. "Trot gait design and CPG method for a quadruped robot." *J. Bionic Eng.* 11 (1): 18–25. https://doi.org/10.1016/S1672-6529(14)60016-0.

Zhao, X., and H. Jebelli. 2025a. "A computational method for real-time roof defect segmentation in robotic inspection." *Comput.-Aided Civ. Infrastruct. Eng.* 40: 3596–3623. https://doi.org/10.1111/MICE.13471.

Zhao, X., and H. Jebelli. 2025b. "Adaptive kinematic and dynamic control for hexapod robots in complex construction environments." In *Proc., 42nd Int. Symp. on Automation and Robotics in Construction*. Montreal, Canada: Concordia University.

Zhao, X., T. Ren, and H. Jebelli. 2024. "Adaptive intelligence for robot navigation efficiency with a deep-reinforcement-learning-based cyber-physical system." In *Proc., 2024 Int. Conf. on Computing in Civil Engineering*. Pittsburgh, PA: Carnegie Mellon University.

Zhao, X., T. Ren, Y. Liu, and H. Jebelli. 2025. "Senser fusion and hierarchical inspection strategy for aerial-ground multi-robot system." In *Proc., 42nd Int. Symp. on Automation and Robotics in Construction*. Montreal, Canada: Concordia University.

Zhao, Y., X. Chai, F. Gao, and C. Qi. 2018. "Obstacle avoidance and motion planning scheme for a hexapod robot Octopus-III." *Rob. Auton. Syst.* 103: 199–212. https://doi.org/10.1016/j.robot.2018.01.007.

ASCE OPEN: Multidiscip. J. Civ. Eng., 2025, 3(1): 04025011